

Approximate Output Power of The TEA2025 Amplifiers

Table 3

Configuration	+ Vee	RLoad	Pmin	Ptyp	Pmax
Single amplifier	9V	4-ohm	1.7W	2.3W	-
Single amplifier	9V	8-ohm	0.7W	1.3W	-
Single amplifier	6V	4-ohm	0.7W	1.0W	-
Single amplifier	6V	8-ohm	0.7W	0.6W	-
Bridge amplifier	9V	8-ohm	0.45W	4.7W	-
Bridge amplifier	6V	4-ohm	-	2.8W	-

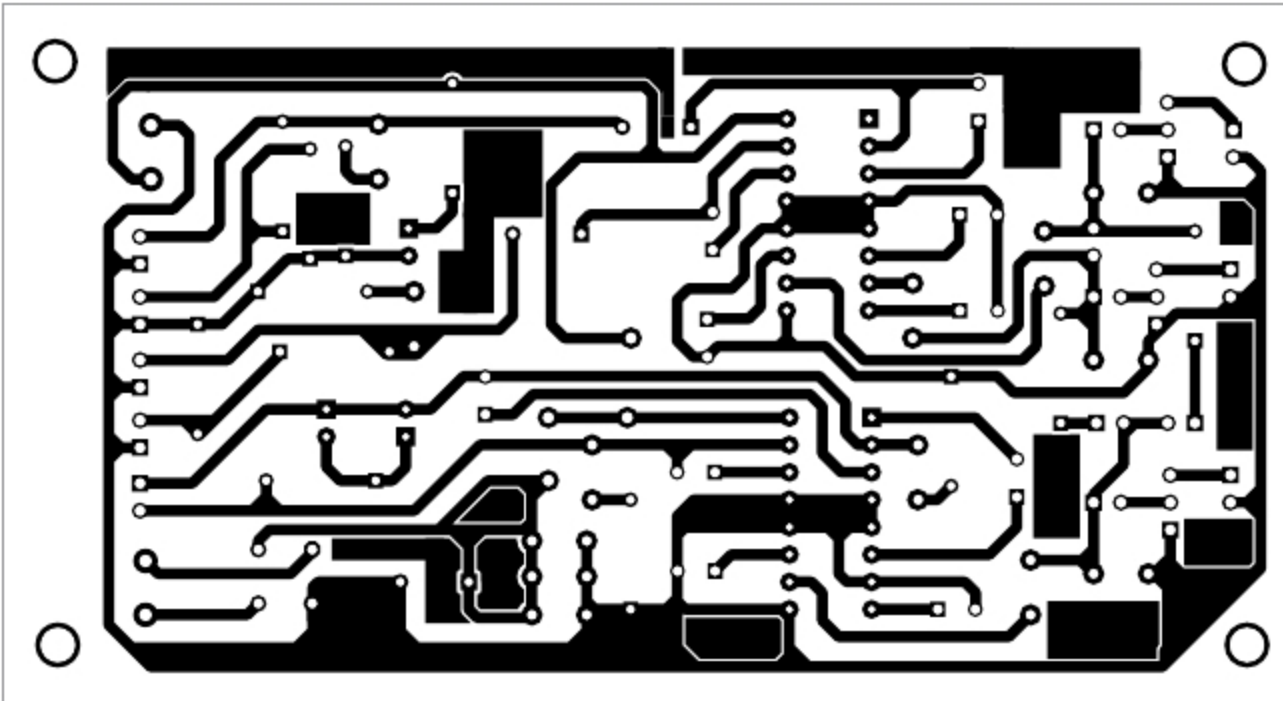


Fig. 2: Actual-size PCB layout for audio amplifier with three channels

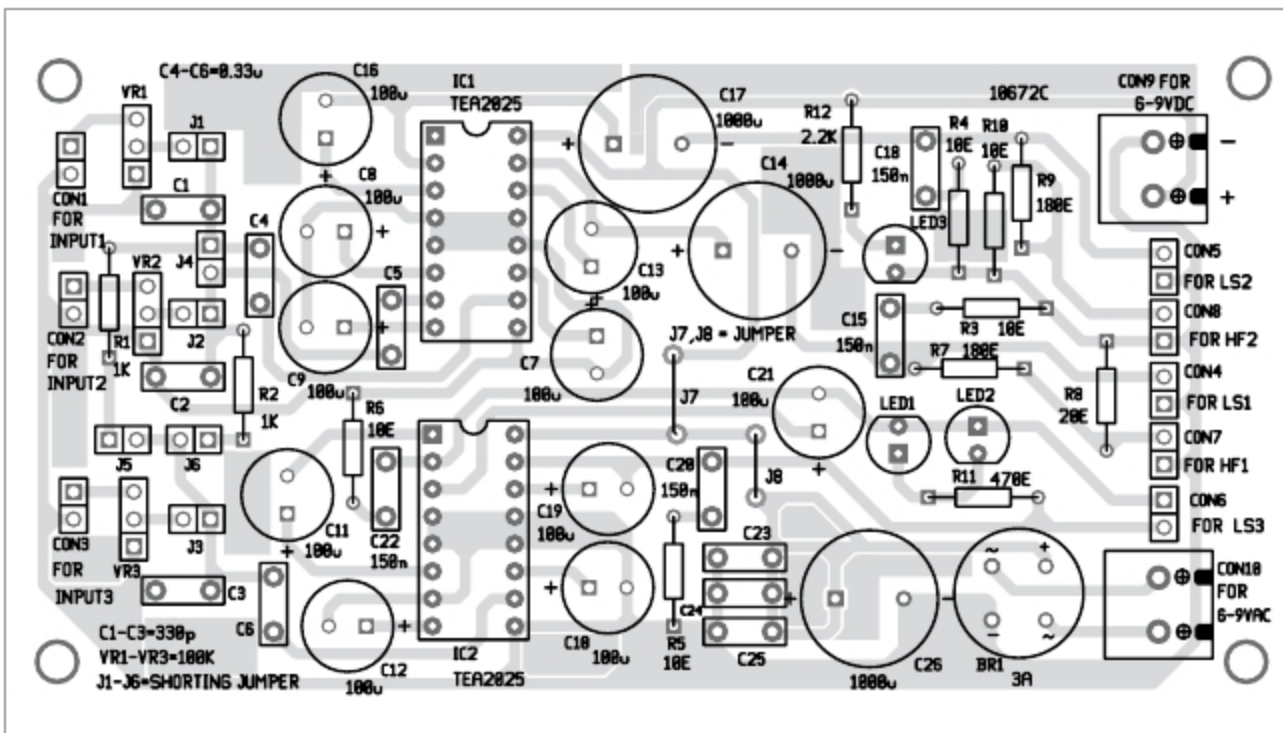


Fig. 3: Components layout of the PCB

audio amplifiers.

The input audio signal is applied to connectors CON1, CON2 and CON3. IC1 works as two separate audio amplifiers, each of them with output power more than 2W. The suggested loads across IC1 are loudspeakers LS1 and LS2 (connected to CON4 and CON5) with a power of 4W or more.

Also, you may use appropriate headphones (HF1 and HF2) with an impedance of 8-ohm or more, connected to CON7 and CON8.

IC2 works as a single bridge amplifier with output power more than 4W. It is suggested to use loudspeaker LS3 (connected to CON6) with a power of 6W or more.

A 6-9V DC voltage can be applied to connector CON9 and 6-9V AC voltage to connector CON10. You can use a 50/60Hz 230V AC primary to 9V rms, 3A secondary transformer X1 for AC supply at CON10. A 3A bridge rectifier BR1 is used across CON10. Alternatively, you can use four 1N540X or similar diodes with at least 3A and 50V ratings in place of BR1.

Amplifier voltage gain

TEA2025 has relatively high voltage gain, which can reach 220dB or 47dB. Approximate voltage gains in the TEA2025 amplifier in different modes of operations are listed in Table 1.

Use external resistors to reduce the gain down to around 36dB. But reducing the gain to less than 36dB will make the amplifier unstable. External resistors can be connected between pins 11 and 6 of IC1 and the cor-

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